Binh Minh Tran

Contact Information

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Education

2027 – 2031 **Doctor of Philosophy, Medical Science**
Computational RNA Biology, John Curtin School of Medical Research, The Australian National University, Canberra, Australia.
PhD supervision and funding confirmed with Professor Eduardo Eyras.

Feb 2023 – Present **Bachelor of Advanced Computing (Honours)**
Computational Data Science, The University of Sydney, Australia.
Expected completion: December 2026. Cumulative WAM: **82/100 (Distinction)**.
Thesis: *Multimodal Breast Cancer Survival Prediction with Hierarchical Niche–Interface Graph Learning.*
Supervisor: Associate Professor Xiuying Wang.

2019 – 2022 **High School Diploma, Specialised Physics Programme**
HUS High School for Gifted Students, VNU University of Science, Vietnam National University, Hanoi, Vietnam.
GPA: **9.5/10**. Outstanding academic achievement award, 2019–2022.

Research Interests

Machine Learning **AI in Medicine** **Computational Biology** **RNA Biology** **Cancer Genomics**

Publications

Applicant's name in bold. † Equal contribution.

Peer-Reviewed Conference Papers

[C4] **B. M. Tran**, L. Zhang, X. Wang. HiNT-Surv: Hierarchical Niche-Aware Multimodal Survival Prediction from Imaging Mass Cytometry, Clinicopathologic Variables, and Genomic Data. *IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM)* at IEEE SSCI, 2027. Accepted (full paper)

[C3] **B. M. Tran**†, V. A. Nguyen†, L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, T. B. V. Nguyen, D. H. Tran. A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction.
International Symposium on Integrated Uncertainty in Knowledge Modelling and Decision Making (IUKM), Springer LNAI vol. 16918, pp. 102–114, 2026. [DOI] [PDF]

[C2] L. H. Nguyen, **B. M. Tran**, D. H. Tran. A Variational Graph Neural Network with Multi-view Integration for Predicting lncRNA–Disease Associations.
Asian Conference on Intelligent Information and Database Systems (ACIIDS), Springer CCIS vol. 2964, pp. 355–369, 2026. [DOI]

Publications — Continued

[C1] L. H. Nguyen, **B. M. Tran**, D. H. Tran. Dual-Stream Contrastive Graph Representation Learning for lncRNA–Disease Association Prediction.
FISU Joint Conference on Artificial Intelligence (FJCAI), pp. 151–158, 2026. [ISBN]

Peer-Reviewed Journal Articles

[J1] T. B. V. Nguyen, N. M. H. Nguyen, **B. M. Tran**, Q. A. Dang, V. A. Nguyen. A Deep Learning Model Supporting Classification and Localisation of Breast Tumour Lesions on Ultrasound.
Journal of Medical Research, Hanoi Medical University, 2026 (in Vietnamese). Accepted

Manuscripts Under Review

[M2] **B. M. Tran**†, V. A. Nguyen†, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, D. H. Tran, T. B. V. Nguyen. Development and Cross-Cohort Evaluation of a Multimodal Deep-Learning Model for ER, PR and HER2 Status Classification in Breast Cancer.
Scientific Reports, 2026. Under review

[M1] **B. M. Tran**, D. Q.-A. Nguyen, M.-H. Ha, D. M. Pham, T. D. Nguyen, Q. B. Doan, D. T. Doan, M.-A. Nguyen, D.-T. Nguyen, O. T.-C. Chen. H2F-GCN: Skeleton-based Hypergraph Frequency Coordinate Fusion Graph Convolution Network for Physical Rehabilitation Assessment.
IEEE Access, 2026. Under review

Research Experience

Nov 2026 (expected)
Incoming Research Intern, Center for Biomedical Informatics, Vingroup Big Data Institute (VinBigdata), Hanoi, Vietnam.

Oct 2026 – Present
Visiting Researcher, Centre for Computational Biomedical Sciences, John Curtin School of Medical Research, The Australian National University, Canberra, Australia.
Host: Prof Eduardo Eyras.

- Conducting preliminary research on AI/ML approaches to interpret genetic variants affecting RNA modifications (m^6A, pseudouridine and m^5C), with a focus on distinguishing predicted modification changes from disease-relevant functional effects.

Feb 2026 – Oct 2026
Honours Research Student, School of Computer Science, The University of Sydney, Australia.
Supervisor: A/Prof Xiuying Wang.

- Designed NIMSurv for interpretable breast-cancer survival prediction using hierarchical graphs, clinical data and imaging mass cytometry.
- Evaluated NIMSurv on 544 METABRIC patients, achieving Uno's C of **0.703** for 10-year disease-specific survival under patient-level cross-validation.
- Developed HiNT-Surv, an earlier model accepted as a full paper at IEEE SSCI 2027 [C4].

Dec 2025 – Present
Research Collaborator, Department of Obstetrics & Gynaecology, Hanoi Medical University, Vietnam.

- Developed a multimodal, multitask model integrating clinical data, H&E whole-slide images and tumour RNA-seq to predict ER, PR and HER2 status in breast cancer [C3].
- Conducted cross-validation on 937 TCGA-BRCA cases and external testing on 122 CPTAC-BRCA cases, achieving an external macro AUC of **0.931** without retraining.

Dec 2025 – Mar 2026
Research Assistant, Center for Biomedical and Pharmaceutical Research, Vietnam Academy of Science and Technology (VAST), Vietnam.

- Applied reproducible NGS workflows for mutation and microbial gene-function analyses: FastQC/Trimmomatic preprocessing, genome assembly, gene prediction and functional annotation.

Research Experience — Continued

Dec 2024 – Dec 2025

Research Collaborator, Hanoi University of Industry, Vietnam.
Supervisor: A/Prof Dang Hung Tran.

- Developed two graph-learning models for lncRNA–disease association prediction: a multi-view variational graph neural network [C2] and a dual-stream contrastive model [C1].
- Evaluated both models and drafted manuscripts, contributing to two peer-reviewed conference publications.

Research Grants

Dec 2026 – Nov 2027

AI Model for Individualised Post-Test Probability after High-Risk NIPT
Department of Obstetrics & Gynaecology, Hanoi Medical University.
Role: Co-principal investigator, with Nguyen Van An.
Supervisor: Nguyen Thi Bich Van, MD, PhD.
Funding: VND 10,000,000 (Hanoi Medical University).
Aim: Develop and calibrate an individualised probability model for confirmed trisomy 21, 18 or 13 following a high-risk NIPT result, using signed Z-score, fetal fraction and assay platform in retrospective high-risk referrals.

Honours and Awards

2026 — **Fourth Prize**, 23rd Youth Science and Technology Conference of the Vietnamese Health Sector
For breast-cancer receptor-status research [C3]; national level.

2026 — **Top-10 Finalist**, AURORA Scientific Research Awards, Vietnam–Australia Innovation Network (NIC-AU)
Grand Final in Hanoi, December 2026.

2026 — **Outstanding Student Award**, Embassy of Vietnam in Australia
For academic and research achievement.

2026 — **Second Prize**, Vietnamese Young Intellectuals Forum in Australia
For breast-cancer receptor-status research [C3].

2026 — **Third Prize**, Youth Science and Technology Conference, Hanoi Medical University
For breast-cancer receptor-status research [C3]; university level.

2025 — **January Star Award**, Vietnam National Union of Students
For academic achievement, research and student leadership; one of 117 national recipients.

2023 — **Sydney International Student Award**, The University of Sydney
Tuition support awarded at admission for the duration of the degree.

Industry Experience

Dec 2024 – Mar 2025 — **Data Engineer Intern**, Viettel Software, Hanoi, Vietnam.
Built and maintained Python/SQL ETL pipelines; contributed to data-quality monitoring, schema design and pipeline/query performance tuning.

Jun – Jul 2024 — **Data Engineer Intern**, Agribank, Hanoi, Vietnam.
Supported ETL into reporting systems and contributed to dashboards and operational reports for internal stakeholders.

Research Skills

Machine learning	Graph neural networks, multimodal fusion, multiple-instance learning, self-supervised and contrastive learning, Cox and discrete-time survival models
Biomedical data	H&E whole-slide images, imaging mass cytometry, clinical data, RNA-seq and gene expression, NGS preprocessing and annotation, lncRNA–disease networks
Evaluation	Patient-level nested cross-validation, bootstrap confidence intervals, paired tests with multiple-testing correction, effect sizes, calibration, external validation
Programming	Python (PyTorch, PyTorch Geometric, scikit-learn), R, SQL, Bash
Tools	Git, Linux, LaTeX

Languages

Vietnamese (native), English (IELTS 7.0), French (DELF A1)

References

Referee 1	**Associate Professor Xiuying Wang** School of Computer Science, The University of Sydney. Email: xiu.wang@sydney.edu.au
Referee 2	**Associate Professor Dang Hung Tran** Hanoi University of Industry. Email: hungtd@haui.edu.vn
Referee 3	**Professor Eduardo Eyras** John Curtin School of Medical Research, ANU. Email: eduardo.eyras@anu.edu.au